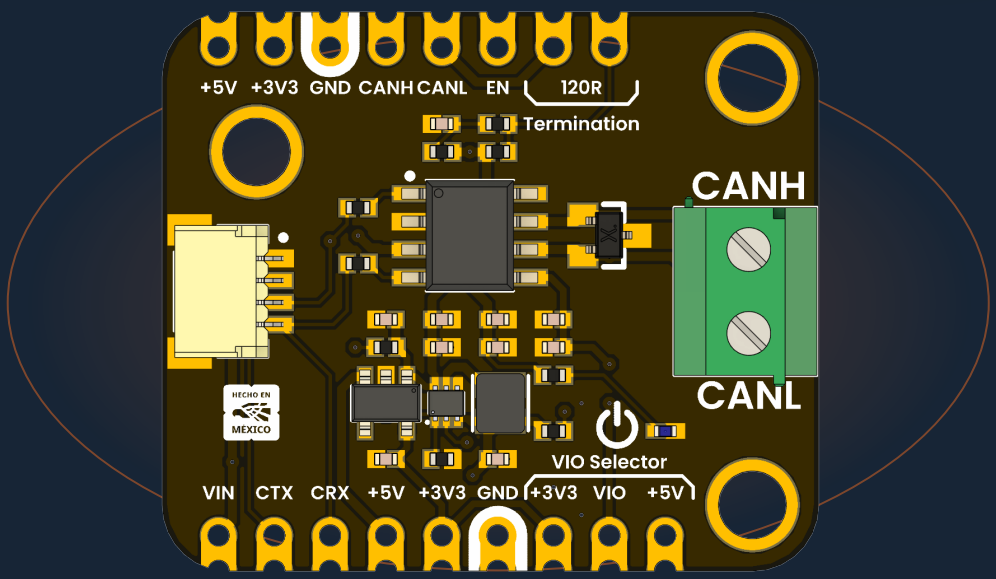
PRODUCT REFERENCE MANUAL

DEVLAB FACTOR

# TCAN1051HVD CAN Transceiver Module

**MPN: UE0085**

Version 1.0.0




Fault-protected high-speed CAN physical-layer interface with selectable 3.3 V or 5 V controller logic.

Classic CAN | CAN FD · 2 Mbps | 3.3 V / 5 V I/O | 120 Ω termination

DOCUMENT MAP

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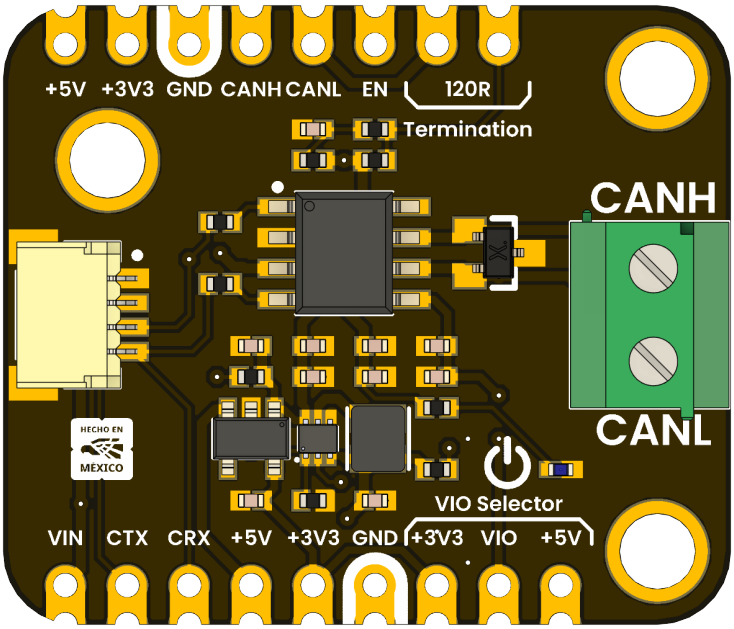
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# Description

The **DevLab TCAN1051HVD CAN Transceiver Module** is a compact physical-layer interface for Classic CAN and CAN FD networks. It connects a microcontroller or stand-alone CAN controller to the differential CAN bus through the CTX (TXD), CRX (RXD), CANH, and CANL signals.

The module is built around the TCAN1051HVD, a fault-protected high-speed CAN transceiver with a separate VIO supply. An onboard boost converter and 3.3 V regulator generate the transceiver and logic rails from VIN. The VIO selector allows direct use with either 3.3 V or 5 V controllers.




> This module implements the CAN physical layer only. The host must provide a CAN controller or a microcontroller with an integrated CAN/TWAI peripheral.

## Applications

- Industrial automation and distributed control
- Robotics and mobile platforms
- Building and environmental control
- Automotive prototyping and diagnostics
- CANopen, DeviceNet, NMEA 2000, ISOBUS, and other CAN-based networks

## Hardware Features

- TCAN1051HVD high-speed, fault-protected CAN transceiver
- Classic CAN physical layer and CAN FD data rates up to 2 Mbps with a compatible controller
- Selectable 3.3 V or 5 V logic interface (VIO)
- Onboard TPS61023 5 V boost converter and AP2112K 3.3 V regulator
- Switchable split 120 Ω termination (2 × 60 Ω with 4.7 nF center capacitor)
- PESD2CANFD24L-UX protection device on CANH and CANL
- 22 Ω series resistors on CTX and CRX
- 3.81 mm screw terminal for the CAN bus
- 4-pin, 1 mm-pitch connector for VIN, GND, CTX, and CRX
- Castellated headers for breadboard, carrier-board, or direct-wire integration
- Power indicator LED and exposed regulator enable signal

# 1 The Board

The DevLab TCAN1051HVD CAN Transceiver Module (UE0085) provides a physical-layer interface between a CAN-capable host controller and a differential CAN bus.

The design separates the controller-side logic interface from the CANH and CANL bus signals. Controller connections are available through a 4-pin JST-compatible connector and castellated contacts, while the CAN bus is accessible through a 3.81 mm screw terminal and dedicated board contacts.

An onboard 5 V boost converter, 3.3 V linear regulator, selectable logic voltage, and CAN bus protection circuitry support integration with 3.3 V and 5 V controllers.

## 1.1 Accessories

The UE0085 module is supplied with the following accessories:

| Accessory | Quantity | Purpose |
|---|---|---|
| DevLab TCAN1051HVD CAN Transceiver Module (UE0085) | 1 | Main CAN physical-layer interface board |
| JST/Qwiic-compatible cable (AR4561) | 1 | Connects the module to the host controller through the 4-pin interface |
| Single-row male header, 1 × 9 pins | 1 | Provides solderable connections for board integration |
| Single-row male header, 1 × 8 pins | 1 | Provides additional solderable connections |
| Jumper bridge, 2.54 mm | 2 | Used for VIO logic-voltage selection (3.3 V or 5 V) and enabling the onboard 120 Ω CAN bus termination |

The supplied headers and jumper bridges may require installation depending on the intended application.

Select only one VIO voltage level at a time. Enable the 120 Ω termination only when the module is installed at a physical end of the CAN bus.

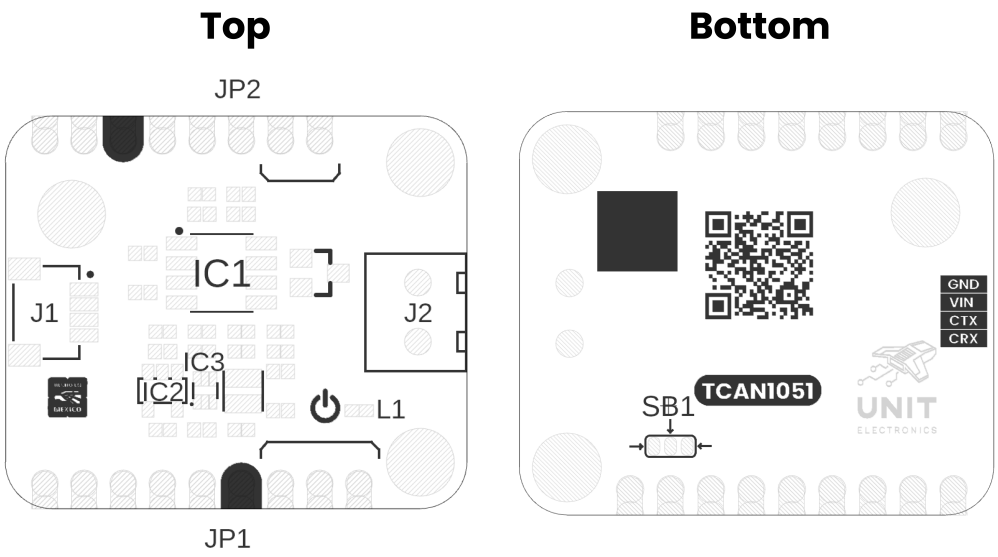
## 1.2 Board Identification

| Item | Value |
|---|---|
| Product | DevLab TCAN1051HVD CAN FD Transceiver Breakout |
| Brand / company | UNIT Electronics |
| Board ecosystem | DevLab |
| Product type | High-speed CAN / CAN FD physical-layer transceiver module |
| Main transceiver | TCAN1051HVD |
| Power converter | TPS61023 |
| Voltage regulator | AP2112K-3.3 |
| Logic interface | Selectable 3.3 V / 5 V |

| Item | Value |
|---|---|
| CAN interface | CANH and CANL differential bus |
| Manufacturer Part Number (MPN) | UE0085 |
| Available schematic | Revision 1 |
| Product Reference | Version 1.0.0 |

Board, schematic, pinout, and documentation revisions are controlled independently.

## 1.3 Board Views




**Views of Board Topology**

### Views of Board Topology

The top view identifies the main integrated circuits, controller and CAN bus connectors, power circuitry, and castellated contacts.

The bottom view shows the VIO solder selector, board identification markings, and controller signal references.

| Reference | Description | Location |
|---|---|---|
| IC1 | TCAN1051HVD CAN transceiver | Top |
| IC2 | AP2112K-3.3 linear regulator | Top |
| IC3 | TPS61023 boost converter | Top |
| J1 | 4-pin controller interface connector | Top |
| J2 | 2-position CAN bus screw terminal | Top |
| JP1 | CAN bus termination selector | Top |
| JP2 | Controller and power castellated contacts | Top |
| L1 | Boost converter inductor | Top |

| Reference | Description | Location |
|---|---|---|
| SB1 | VIO solder selector | Bottom |
| QR code | Board identification and product reference | Bottom |
| GND, VIN, CTX, CRX | Controller interface signal markings | Bottom |

## 1.4 Handling

Use standard electrostatic discharge (ESD) precautions when handling or installing the module.

Remove power before installing headers, changing jumper positions, or modifying the VIO selection.

Select only one VIO voltage level at a time. Connecting the 3.3 V and 5 V rails together may damage the module or the connected host.

Verify CANH and CANL polarity before connecting the module to an active CAN network.

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus. Intermediate nodes must not introduce additional termination resistance.

Do not connect external power directly to the regulated 3V3 or 5V outputs unless the application has been specifically validated for that configuration.

The module does not provide galvanic isolation between the host controller and the CAN bus.

# 2 Ratings and Electrical Characteristics

## 2.1 Recommended Operating Conditions

| Symbol | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VIN | Module input voltage | 3.3 | — | 5.0 | V |
| VCC | TCAN1051HVD bus supply | 4.5 | 5.0 | 5.5 | V |
| VIO | Selected I/O supply | 2.8 | 3.3 / 5.0 | 5.5 | V |
| VOUT(5V) | TPS61023 regulated output, nominal | — | 5.0 | — | V |
| VOUT(3V3) | AP2112K regulated output, nominal | — | 3.3 | — | V |
| IO(RXD) | Recommended CRX output-current range | −2 | — | 2 | mA |
| TA | TCAN1051HVD operating free-air temperature | −55 | — | 125 | °C |
| CAN FD | Supported data rate for TCAN1051HVD | — | — | 2 | Mbps |

VIN represents the recommended module input voltage. VCC, VIO, I/O current, temperature, and CAN FD values are TCAN1051HVD specifications. The 1 Mbps limit of the ESP32-H2/P4 example is a controller limit, not a transceiver limit.

The 5 V and 3.3 V rails are generated by the onboard TPS61023 boost converter and AP2112K-3.3 linear regulator, respectively.

The complete assembled module has not been independently qualified over the full IC temperature range.

## 2.2 Selected Transceiver Characteristics

Unless otherwise noted, the following characteristics apply to the TCAN1051HVD transceiver.

| Symbol | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VIH | CTX high-level input voltage | 0.7 × VIO | — | — | V |
| VIL | CTX low-level input voltage | — | — | 0.3 × VIO | V |
| VOH | CRX high-level output voltage at −2 mA | 0.8 × VIO | — | — | V |
| VOL | CRX low-level output voltage at 2 mA | — | — | 0.2 × VIO | V |
| VO(DOM), CANH | Dominant CANH output voltage | 2.75 | — | 4.5 | V |
| VO(DOM), CANL | Dominant CANL output voltage | 0.5 | — | 2.25 | V |

| Symbol | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VOD(DOM) | Dominant differential output, 50–65 Ω load | 1.5 | — | 3.0 | V |
| VCM | Receiver common-mode range | −30 | — | 30 | V |
| tPROP(LOOP) | Total loop delay | — | 100 / 110 | 160 / 175 | ns |

## 2.3 Power Regulation Characteristics

The module integrates a TPS61023 boost converter and an AP2112K-3.3 low-dropout linear regulator.

### TPS61023 — 5 V Boost Converter

| Symbol | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VIN(IC) | Converter operating input voltage | 0.5 | — | 5.5 | V |
| VSTART | Converter startup input voltage | — | — | 1.8 | V |
| VOUT | Adjustable output voltage range | 2.2 | — | 5.5 | V |
| VOUT(MOD) | Configured module output voltage | — | 5.0 | — | V |
| fSW | Switching frequency, VIN > 1.5 V | — | 1.0 | — | MHz |
| ILIM | Valley switch current limit, typical | — | 1 | — | A |

The TPS61023 generates the regulated 5 V rail from VIN.

The converter current-limit specification refers to the internal switching current and must not be interpreted as the available module output current.

The maximum external load on the 5 V rail has not been independently characterized.

### AP2112K-3.3 — 3.3 V Linear Regulator

| Symbol | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VIN(LDO) | Regulator input voltage, nominal | — | 5.0 | — | V |
| VOUT | Fixed regulated output voltage, nominal | — | 3.3 | — | V |
| IOUT | Rated regulator output-current capability | — | — | 600 | mA |
| VDROP | Dropout voltage | TBD | TBD | TBD | mV |
| TA | Regulator operating temperature | TBD | — | TBD | °C |

The AP2112K-3.3 generates the module's 3.3 V rail from the regulated 5 V supply.

The 600 mA rating represents the regulator IC capability, not the guaranteed external current available from the assembled module.

Available output current depends on regulator thermal dissipation, PCB layout, input voltage, and total module power consumption.

Dropout voltage and operating-temperature limits require confirmation against the exact AP2112K manufacturer datasheet.

## 2.4 Absolute Maximum Ratings

Absolute maximum ratings are stress limits, not normal operating conditions.

### TCAN1051HVD — CAN Transceiver

| Symbol | Description | Min | Max | Unit |
|---|---|---|---|---|
| VCC | TCAN1051HVD 5 V supply | −0.3 | 7 | V |
| VIO | TCAN1051HVD I/O supply | −0.3 | 7 | V |
| VBUS | CANH or CANL, H-suffix device | −70 | 70 | V |
| VDIFF | CANH-to-CANL differential voltage, H-suffix device | −70 | 70 | V |
| IO(RXD) | CRX output current | −8 | 8 | mA |
| TJ | IC junction temperature | −55 | 150 | °C |
| TSTG | IC storage temperature | −65 | 150 | °C |

The ±70 V bus-fault rating applies only to the specified CAN bus pins and must not be applied to VIN, CTX, CRX, EN, VIO, 3V3, or 5V.

### Power Regulators

The TPS61023 and AP2112K-3.3 have independent absolute maximum ratings defined by their respective manufacturers.

Their absolute maximum input, output, and temperature ratings must not be interpreted as the operating limits of the complete UE0085 module.

## 2.5 Module Power and Thermal Considerations

The module uses the following power architecture:

- VIN supplies the TPS61023 boost converter.
- TPS61023 generates the regulated 5 V rail.
- AP2112K-3.3 generates the 3.3 V rail from 5 V.
- TCAN1051HVD operates from the regulated 5 V supply.
- VIO selects either the 3.3 V or 5 V logic rail.
- Both regulated rails are accessible through the module headers.

The AP2112K power dissipation can be estimated as:

$P_D \approx (V_{IN(LDO)} - V_{OUT}) \times I_{OUT}$

At a 5 V regulator input and 3.3 V output, the voltage difference is approximately 1.7 V.

The available external current on the 3.3 V and 5 V rails has not been independently qualified.

Do not exceed the thermal, electrical, or current limits of any individual component.

# 3 Functional Overview

## 3.1 Signal Path

The TCAN1051HVD TXD input is driven by CTX. A dominant bus state is requested by a logic LOW, while a recessive bus state is requested by a logic HIGH. The transceiver transforms this logic signal into the CANH/CANL differential output. The TCAN1051HVD RXD output drives CRX. If the receiver detects a dominant bus state, CRX is LOW; if it detects a recessive bus state, CRX is HIGH. The CRX output level is determined by the selected VIO rail.

The TCAN1051HVD is equipped with thermal shutdown, dominant-timeout, and undervoltage protection. When the transceiver is unpowered, the bus and logic pins exhibit a high impedance.

## 3.2 Power Architecture

VIN feeds the TPS61023 boost converter, which generates the 5 V transceiver supply. The AP2112K then generates the 3.3 V rail. Both regulated rails are exposed on the castellated headers.

The VIO selector connects the transceiver I/O supply to either 3.3 V or 5 V:

| Host logic | VIO selector |
|---|---|
| 3.3 V MCU or CAN controller | 3V3 |
| 5 V MCU or CAN controller | 5V |

Select exactly one side of the VIO solder selector. Bridging both sides shorts the 3.3 V and 5 V rails and can damage the module or host.

EN controls the TPS61023 regulator. The onboard bias enables the regulator for normal use; driving EN LOW disables the generated supply rails and transceiver.

## 3.3 Bus Termination

The onboard termination is a split 120 Ω network made from two 60 Ω resistors. Its midpoint is AC-coupled to ground by 4.7 nF to reduce common-mode noise.

Close the **120R Termination** selector only when this module is installed at one physical end of the CAN trunk. A correctly wired point-to-point bus normally has one 120 Ω terminator at each end (approximately 60 Ω measured between CANH and CANL with power removed). Leave the selector open at intermediate nodes.

When the selector is closed, the external termination remains connected even if IC1 is unpowered. The IC's high-impedance, unpowered bus behavior does not remove that board-level 120 Ω load.

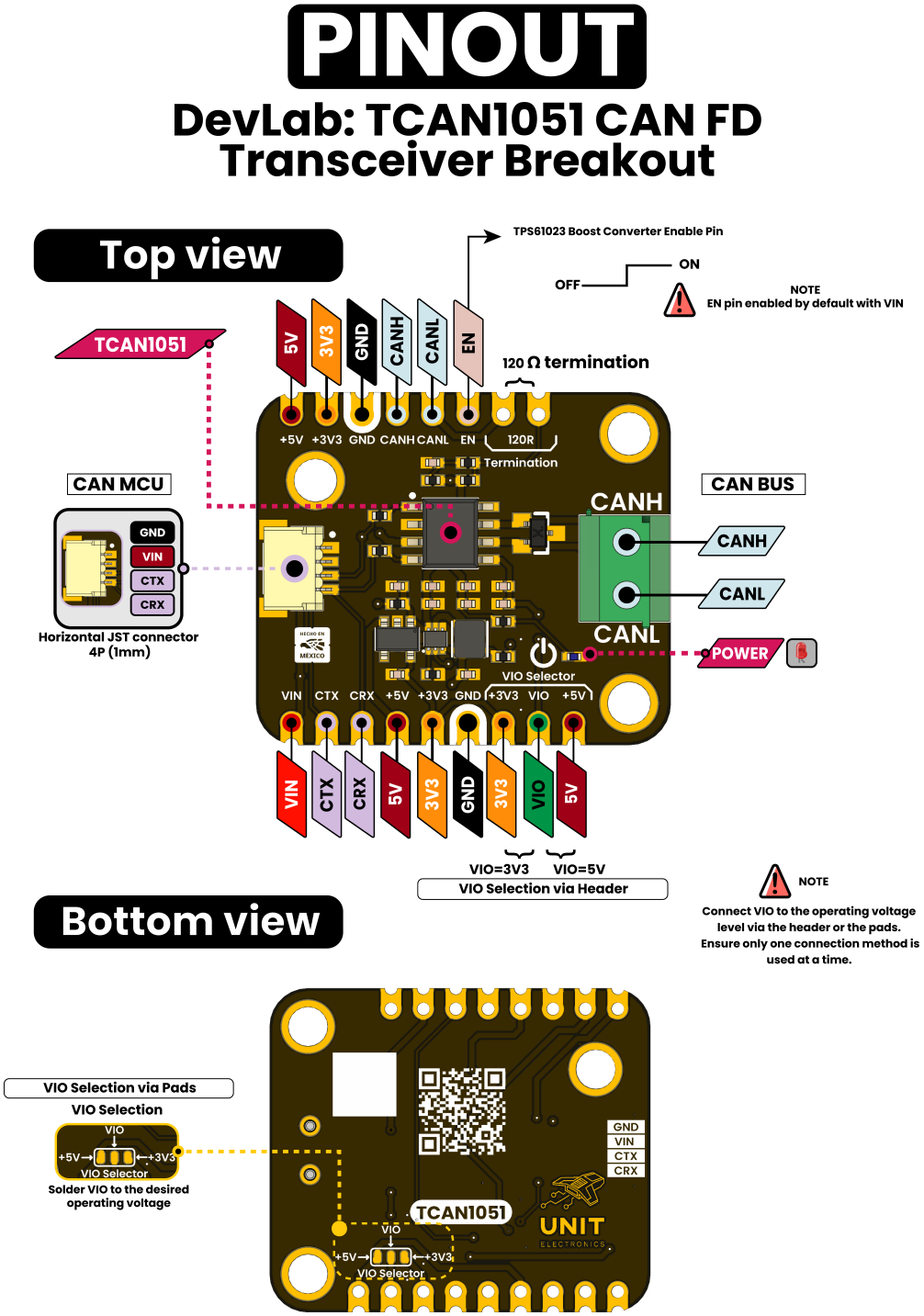
## 3.4 Bus Protection

CANH and CANL are protected by a PESD2CANFD24L-UX device. The TCAN1051HVD H-suffix transceiver provides a ±70 V bus-fault rating and a ±30 V receiver common-mode range. These protections improve robustness but do not replace correct grounding, cabling, termination, or system-level surge design.

# 4 Connectors and Pinouts

The UE0085 module provides controller-side logic connections, differential CAN bus terminals, and castellated contacts for integration with external hardware.

The following pinout identifies the available signals, power connections, and onboard configuration options.



## 4.1 Signal Reference

| Label | Direction | Description |
|---|---|---|
| VIN | Power input | Module input supply, 3.3 V to 5.0 V |
| GND | Power | Common module and bus reference |
| CTX | Input | Controller transmit signal; LOW = dominant |
| CRX | Output | Controller receive signal; LOW = dominant |
| CANH | Bus I/O | CAN high bus line |
| CANL | Bus I/O | CAN low bus line |
| EN | Input | 5 V boost-converter enable; LOW disables the generated supply rails |
| 5V | Power output | Generated 5 V transceiver rail |
| 3V3 | Power output | Generated 3.3 V logic rail |
| VIO | Power | Selected transceiver logic supply (3.3 V or 5 V) |

The CTX and CRX signals operate at the voltage level selected through VIO. The host controller must use compatible logic levels. The module implements the CAN physical layer only. An external CAN controller or a microcontroller with an integrated CAN peripheral is required.

## 4.2 J1 Controller Connector

J1 is a 4-position, 1 mm-pitch connector providing power and communication signals to the host controller.

Use the bottom-side silkscreen to confirm the pin order before connecting a cable.

| Signal | Connection |
|---|---|
| GND | Host ground |
| VIN | Host power supply |
| CTX | Host CAN TX output |
| CRX | Host CAN RX input |

Verify the cable orientation and signal order before connecting the module to an external controller.

## 4.3 J2 CAN Bus Terminal

J2 is a 2-position, 3.81 mm screw terminal providing the differential CAN bus connections.

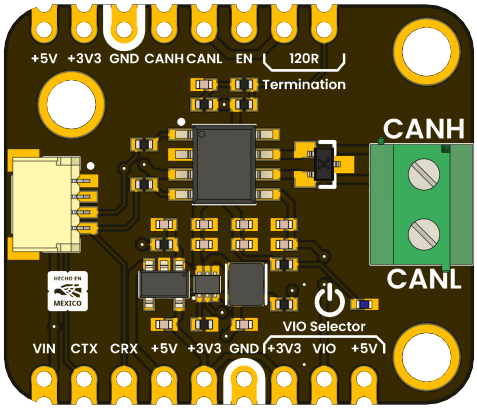
| Position | Signal | Connection |
|---|---|---|
| Upper | CANH | CAN bus high line |
| Lower | CANL | CAN bus low line |

Use twisted-pair wiring for CANH and CANL. Maintain correct bus polarity and minimize stub lengths to preserve signal integrity.

## 4.4 Castellated Headers

The castellated contacts provide access to controller signals, CAN bus connections, regulated power rails, and configuration signals.

These contacts support direct wiring, carrier-board integration, and prototyping applications.




The contact labels are printed on the board silkscreen. Refer to the pinout illustration for signal identification.

## 4.5 Configuration Jumpers

The module includes two jumper-selectable functions: VIO logic-voltage selection and CAN bus termination.

| Configuration | Selection | Function |
|---|---|---|
| VIO | 3V3 | Sets the transceiver logic interface to 3.3 V |
| VIO | 5V | Sets the transceiver logic interface to 5 V |
| CAN termination | Enabled | Connects the onboard 120 Ω termination network |
| CAN termination | Disabled | Disconnects the selectable termination network |

### VIO Selection

Select the VIO voltage according to the logic level of the connected host controller.

Only one VIO voltage selection must be active at a time. Connecting both the 3.3 V and 5 V rails together may damage the module or the host controller.

### CAN Bus Termination

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus.

For intermediate nodes, leave the termination disabled.

A correctly terminated CAN bus normally uses one 120 Ω termination at each physical end of the main bus.

The termination network and jumper configuration must match the released hardware assembly revision.

# 5 Board Operation

This procedure brings up a two-node, 500 kbit/s Classic CAN bus. It uses one UE0085 module with a Pulsar ESP32-H2 transmitter and a second UE0085 module with an ESP32-P4 receiver. The module supplies the CAN physical layer; each ESP32 supplies the TWAI/CAN controller.

## 5.1 Getting Started with Arduino IDE

Gather the hardware below before configuring Arduino IDE and uploading the example sketches.

| Item | Quantity | Purpose |
|---|---|---|
| UE0085 TCAN1051HVD module | 2 | One transceiver per controller |
| Pulsar ESP32-H2 and ESP32-P4 development boards | 1 each | TWAI controllers |
| USB data cables and a computer | 2 | Uploads and serial monitoring |
| Twisted pair, hookup wires, common ground wire | As needed | Bus and host wiring |
| Regulated 3.3 V to 5.0 V supply for each module | 1 per module | UE0085 VIN; a host 3.3 V rail is suitable only if it can supply the module |
| Multimeter | Recommended | Power and termination checks |

Install Arduino IDE, an ESP32 board package supporting the selected boards, and the DevLab_TCAN1051HVD Arduino library. This library wraps the ESP32 TWAI peripheral; it does not add a CAN controller to a host without one. Confirm GPIO availability against the actual development-board revisions.

## 5.2 Hardware Connection and Configuration

**Remove power before wiring or changing selectors.** Each host needs its own UE0085 module. Connect J1 as follows; these GPIO assignments match the supplied example sketches.

| UE0085 signal | Pulsar ESP32-H2 transmitter | ESP32-P4 receiver |
|---|---|---|
| VIN | Regulated 3.3 V to 5.0 V supply | Regulated 3.3 V to 5.0 V supply |
| GND | Host GND | Host GND |
| CTX | TWAI TX, GPIO12 | TWAI TX, GPIO20 |
| CRX | TWAI RX, GPIO22 | TWAI RX, GPIO21 |

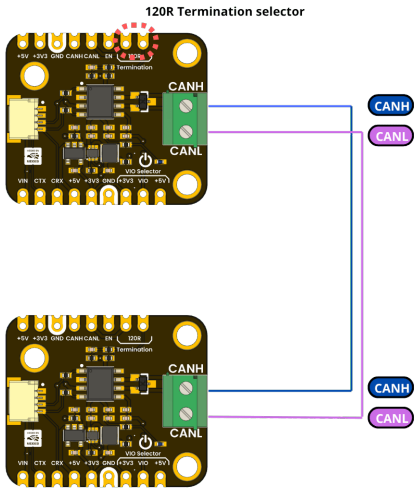
The controller-facing labels belong to the **UE0085 board**. The mounted TCAN1051HVD (IC1) uses different pin names:

| IC1 pin | UE0085 connection | Function |
|---|---|---|
| 1 TXD | CTX through R4 (22 Ω) | Logic input from the controller's CAN TX |
| 2 GND | GND | Common reference |

| IC1 pin | UE0085 connection | Function |
|---|---|---|
| 3 VCC | Onboard regulated 5V rail | Transceiver bus-side supply; not the module's VIN |
| 4 RXD | CRX through R5 (22 Ω) | Logic output to the controller's CAN RX |
| 5 VIO | 3V3/5V selector | Sets TXD/S input thresholds and RXD HIGH level |
| 6 CANL, 7 CANH | J2 bus terminal, with onboard protection and selectable termination | Differential CAN bus |
| 8 S | Unconnected on the revision 1 schematic | Internal pull-down selects normal transceiver mode by default; no external S control |

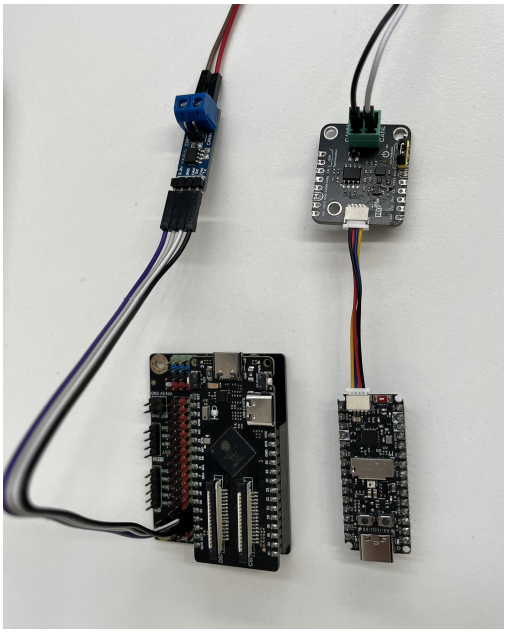
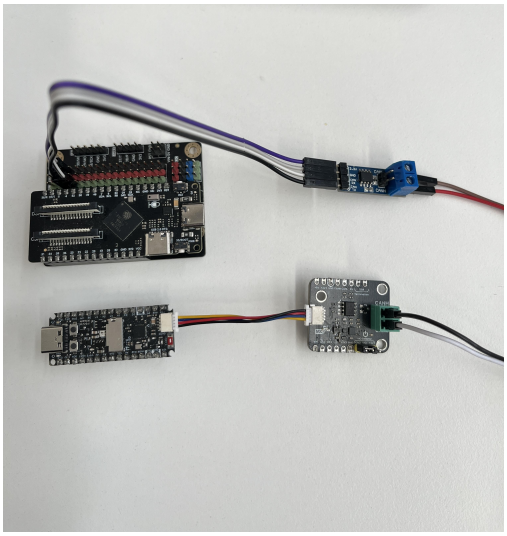
VIN is the **module input**. The labeled 5V and 3V3 contacts are generated output rails, not substitutes for VIN. Set each module's VIO selector to **3V3** for these 3.3 V ESP32 GPIOs. Select exactly one VIO position; bridging 3V3 and 5V shorts the rails. Leave EN in its normal, onboard-enabled state. EN controls the boost converter; it does not select IC1's silent mode. For a 5 V host, select 5V only when its controller logic is 5 V compatible.

Connect the two J2 terminals **CANH to CANH** and **CANL to CANL** with a short twisted pair. Connect both module/host grounds as a common reference, even when the modules have separate supplies. Do not connect CANH or CANL to GPIOs. For this two-node bus, close the **120R Termination** selector on each module: the modules occupy the two physical ends. Do not add external terminators when both onboard networks are enabled. On a larger bus, terminate only its two physical ends and keep branch stubs short.



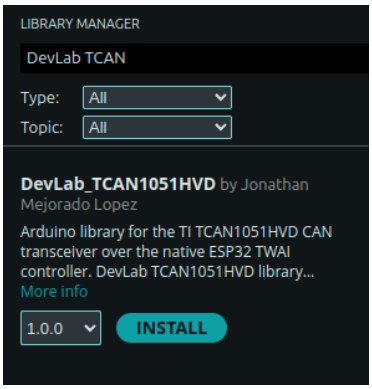

With all power removed, approximately **60 Ω** should measure between CANH and CANL on the assembled two-node bus. Approximately 120 Ω suggests a missing terminator; a much lower reading suggests extra termination or a short. After applying power, check that both green power LEDs illuminate and the 5V and 3V3 rails are near their nominal voltages. Remove power if a rail is shorted, the device heats unexpectedly, or the VIO selector bridges both rails.

The photographs below show an **illustrative mixed-transceiver bench setup**: the black UE0085 is connected to the ESP32-H2, while a different blue CAN transceiver is connected to the ESP32-P4. They show the physical arrangement, not the exact two-UE0085 wiring specified above. If using the blue module, follow its own pinout and supply requirements. In either setup, match CANH, CANL, ground reference, bit rate, and the two physical end terminations.

| Connection overview | Closer view of the two CAN nodes |
| --- | --- |
|  |  |

## 5.3 Arduino Library and Controller Setup

1. Install a compatible ESP32 board package in Arduino IDE. Select the exact ESP32-H2 or ESP32-P4 board and its serial port before each upload.
2. Install `DevLab_TCAN1051HVD` through Library Manager if listed. Otherwise download its repository as a ZIP and use **Sketch > Include Library > Add .ZIP Library**. Restart Arduino IDE if the header does not appear.
3. Open the two sketches in Section 5.4. The library also contains `CAN_Transmitter` and `CAN_Receiver` examples; this repository has `software/examples/basic_tx.ino` and `basic_rx.ino`.




*Arduino Library Manager listing for DevLab_TCAN1051HVD; the displayed version may differ from the installed release*

Both sketches explicitly use **500 kbit/s** and `TWAI_MODE_NORMAL`. Every node must use the same nominal bit rate. Normal mode requires another active node to acknowledge a transmitted frame. The example uses standard 11-bit identifier `0x100` and one data byte. The ESP32-H2 and ESP32-P4 TWAI controllers support Classic CAN up to 1 Mbps and cannot process CAN FD frames. The UE0085's TCAN1051HVD physical layer supports CAN FD up to 2 Mbps when paired with a CAN FD controller. This example uses **Classic CAN frames** at 500 kbit/s. `TWAI_MODE_NORMAL` selects the ESP32 controller mode; it does not drive IC1's unconnected S pin.

## 5.4 CAN Communication Example

Upload this transmitter sketch to the Pulsar ESP32-H2. It queues a frame containing an incrementing byte approximately once per second.

```
#include <DevLab_TCAN1051HVD.h>

DevLab_TCAN1051HVD can(GPIO_NUM_12, GPIO_NUM_22);
uint8_t counter = 0;

void setup() {
  Serial.begin(115200);
  if (!can.begin(TWAI_TIMING_CONFIG_500KBITS(), TWAI_MODE_NORMAL)) {
    Serial.println("CAN initialization failed");
    while (true) delay(1000);
  }
  Serial.println("Transmitter ready");
}

void loop() {
  uint8_t data[1] = {counter};
  if (can.send(0x100, data, 1, false, false, 1000)) {
    Serial.print("Frame queued, counter: ");
    Serial.println(counter);
    counter++;
  } else {
    Serial.println("CAN transmit queue failed");
  }
  delay(1000);
}
```

Upload this receiver sketch to the ESP32-P4. It prints the identifier and counter received from the bus.

```
#include <DevLab_TCAN1051HVD.h>

DevLab_TCAN1051HVD can(GPIO_NUM_20, GPIO_NUM_21);

void setup() {
  Serial.begin(115200);
  if (!can.begin(TWAI_TIMING_CONFIG_500KBITS(), TWAI_MODE_NORMAL)) {
    Serial.println("CAN initialization failed");
    while (true) delay(1000);
  }
  Serial.println("Receiver ready");
}

void loop() {
  TCAN1051_Frame frame;
  if (can.receive(frame, 100) && !frame.extended && !frame.rtr &&
      frame.id == 0x100 && frame.length == 1) {
    Serial.print("ID: 0x");
    Serial.print(frame.id, HEX);
    Serial.print("  Counter: ");
    Serial.println(frame.data[0]);
  }
}
```

The library's `send()` return value means the ESP32 TWAI driver accepted the frame into its transmit queue. By itself, it does **not** prove another node acknowledged or received the frame. The receiver output is the end-to-end communication check.

## 5.5 Upload and Communication Verification

1. With power off, recheck VIN, GND, CTX/CRX, CANH/CANL, both VIO selectors, and the two end terminators. Confirm about 60 Ω across CANH and CANL.
2. Power both boards and modules. Confirm both power LEDs illuminate.
3. Select the ESP32-P4 board and port, upload the receiver, and open its serial monitor at **115200 baud**. Wait for `Receiver ready`.
4. Select the Pulsar ESP32-H2 board and port, upload the transmitter, and open its serial monitor at **115200 baud**. Use separate IDE windows or serial terminals to watch both ports.
5. Confirm that the receiver repeatedly shows `ID: 0x100` and a counter advancing from 0, wrapping after 255. The transmitter should report queued frames at about one-second intervals.

Example receiver output (the library may also print an initialization line):

```
Receiver ready
ID: 0x100  Counter: 0
ID: 0x100  Counter: 1
ID: 0x100  Counter: 2
```

If only the transmitter reports queued frames, communication has not been verified. Check the receiver and bus using Section 5.6.

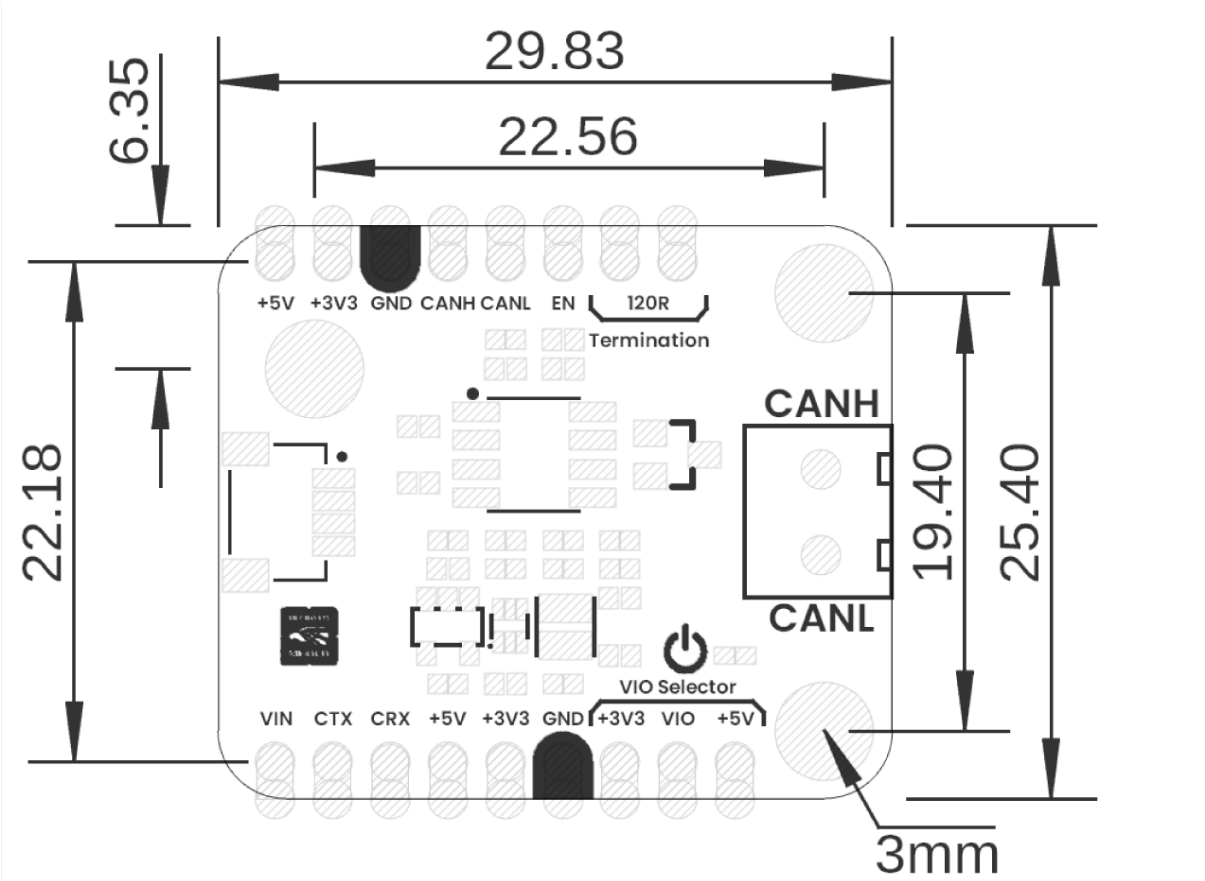
## 5.6 Troubleshooting

| Symptom | Check |
|---|---|
| No power LED or incorrect rail voltage | VIN 3.3 V to 5.0 V, GND continuity, supply capacity, EN not held LOW, and no rail short. |
| Library header missing or compilation fails | Install `DevLab_TCAN1051HVD`, select a compatible ESP32 board package, and confirm `driver/twai.h` is available. |
| `CAN initialization failed` | Board selection, GPIO availability, TWAI support, and whether another sketch or driver has claimed TWAI. |
| Transmitter queues frames but receiver prints nothing | Receiver running, CTX/TX and CRX/RX assignments, CANH/CANL polarity, common ground, 500 kbit/s at both nodes, and 3V3 VIO. |
| Bus errors or repeated retransmission | Second active node in normal mode, one 120 Ω terminator at each end (about 60 Ω with power off), and matching bit rates. `send()` success alone is not an ACK indication. |
| Intermittent frames | Twisted pair, short stubs, sound terminals, ground reference, and bus length appropriate to the bit rate. |
| CTX stays LOW and the node stops transmitting | Check the TX GPIO configuration. IC1's TXD dominant timeout releases its bus driver; returning CTX HIGH clears that condition. Also inspect bus shorts and VIO selection. |

For persistent faults, the library's `getStatus()` API can report TWAI state and bus-error counters. Correct wiring and termination before attempting bus-off recovery.

# 6 Mechanical Information

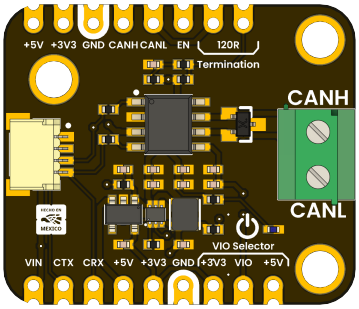
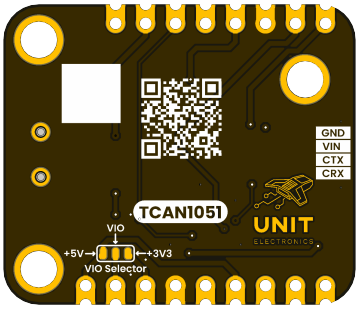
The nominal board outline is **29.83 mm × 25.40 mm**. The drawing below also identifies the 3 mm mounting holes and principal spacing dimensions. All values are in millimeters.




**Board Dimensions in Millimeters**

Use the released mechanical drawing or a physical sample when designing mating enclosures or production fixtures; the rendered image is not a 1:1 fabrication template.

## 6.1 Board Views

| Top | Bottom |
| --- | --- |
|  |  |

# 7 Company Information

| Company name | UNIT Electronics |
|---|---|
| Company website | https://uelectronics.com/ |
| Company Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX |

# 8 Reference Documentation

| Reference | Link |
|---|---|
| Product repository | UNIT-Electronics-MX/ unit_devlab_tcan1051hvd_can_transceiver_module |
| Arduino library | UNIT-Electronics-MX/unit_devlab_tcan1051hvd_library |
| Product Wiki | UNIT Electronics Wiki — UE0085 |
| TCAN1051HV datasheet | SLLSES8E |
| ESP32-H2 datasheet | ESP32-H2 Series |
| ESP32-P4 datasheet | ESP32-P4 Series |
| ISO 11898-2 overview | Refer to ISO 11898-2:2016 for the high-speed CAN physical layer |

The released schematic, manufacturer datasheets, and hardware resources are available in the product repository under `hardware/` and `hardware/resources/datasheet/`.

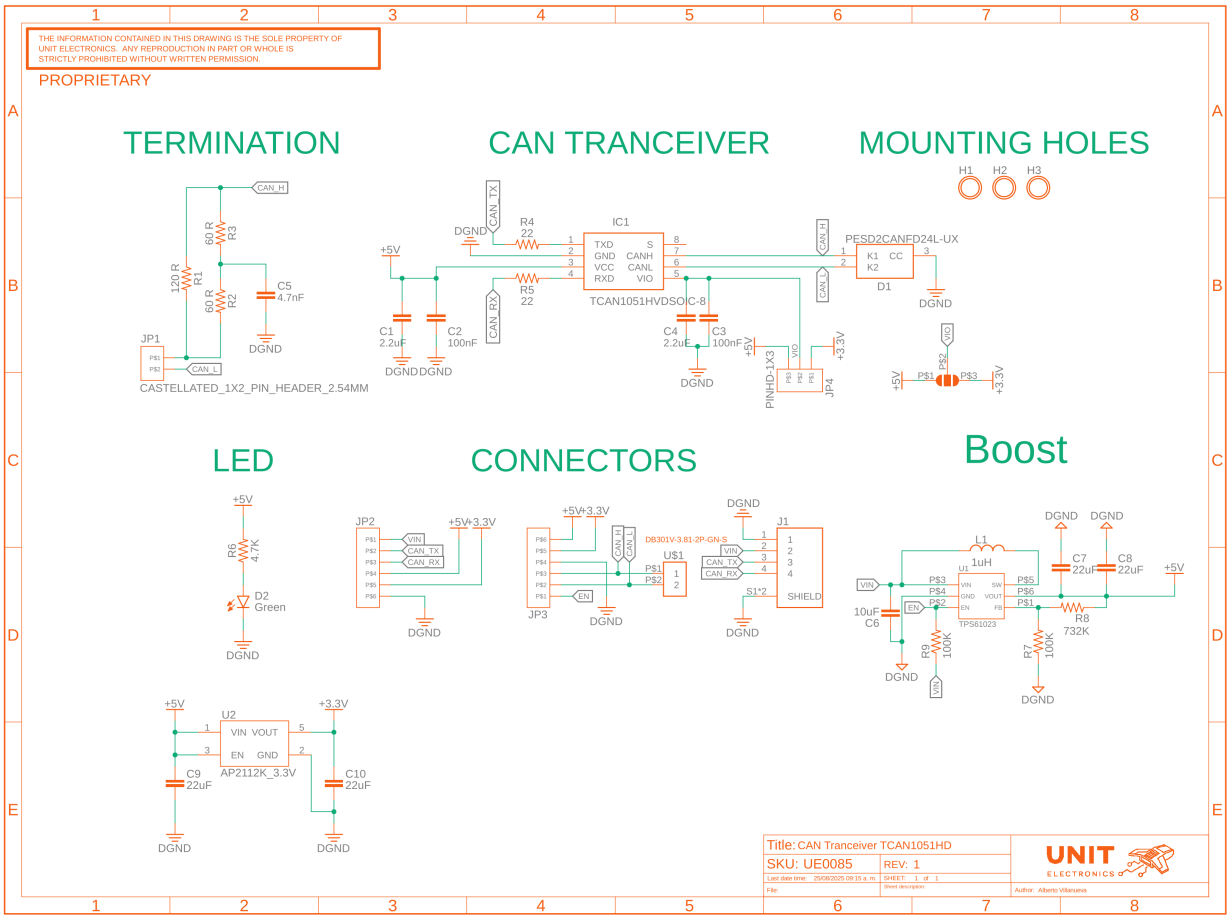
The UNIT Electronics Wiki provides additional product information and integration references.

# 9 Appendix

## 9.1 Schematic

The complete revision 1 schematic is distributed with this product reference:

Open the UE0085 TCAN1051HVD module schematic




*UE0085 revision 1 schematic, also available as the original PDF*

## 9.2 Design Notes

- R1 provides the switchable 120 Ω termination.
- R2 and R3 form the 2 × 60 Ω split termination.
- C5 (4.7 nF) AC-couples the split-termination midpoint to ground.
- R4 and R5 (22 Ω) are in series with CTX and CRX.
- D1 protects CANH and CANL.
- IC1 pin S is unconnected in the revision 1 schematic. The TCAN1051HVD's internal pull-down selects normal mode by default; the module does not expose S as a user control. For noisy or transient-prone systems, review this reliance on the IC's internal bias during hardware qualification.